2022A F=ma Exam: Problem 17

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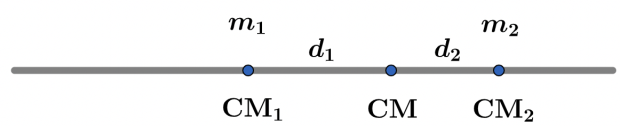

Let the mass of the left (right) side of the rod CM be m_1 (m_2) and the distance between the CM of the left (right) side and the rod CM be d_1 (d_2). Then by definition of center of mass,

$$m_1 d_1 = m_2 d_2$$

Going through the answer choices:

A) $m_1 \neq m_2$ in general so the two parts don't need to have the same mass.

B) The linear momentum of the left side is $p_1 = m_1 v_{\mathrm{CM}_1} = m_1 d_1 \omega$. The linear momentum of the right side is $p_2 = m_2 v_{\mathrm{CM}_2} = m_2 d_2 \omega$. We have $p_1 = p_2$ so this is true.

C) The angular momentum of the left side is $L_1 = I_1 \omega$. The angular momentum of the right side is $L_2 = I_2 \omega$. $I_1 \neq I_2$ in general so the two parts don't need to have the same angular momentum.

D) The kinetic energy of the left side is $K_1 = \frac{1}{2} I_1 \omega^2$. The kinetic energy of the right side is $K_2 = \frac{1}{2} I_2 \omega^2$. $I_1 \neq I_2$ in general so the two parts don't need to have the same kinetic energy.

E) Only one of the above is true so this is incorrect.

Thus, the answer is $\boxed{\text{B}}$.